



Assessment Cover Sheet

Unit number:	ICT307	Unit name:	ICT307 Project 1 (Analysis and Design)
Lecturer or Tutor name:	Dr Nam Hoai Chu		

Student ID	Student name	Contribution
CIHE240495	Yubaraj Thakulla	System architecture, user interface and usability testing
CIHE251125	Nicholas Anthony De Leon	Ethical Issues, Security Threat Assessment, and Usability
CIHE240777	Sangay Yuden	User interface, database design, and usability testing
CIHE240467	Nayan Giri	Ethical Security Considerations, User Interface Design

Submission Title:	System Design Report				
Word Count:	2800	Due date:	24/05/2026	Date submitted:	24/05/2026

- I hold a copy of this assignment if the original is lost or damaged.
- I hereby certify that no part of this assessment or product has been copied from any other student's work or from any other source except where due acknowledgement is made in the assessment.
- I hereby certify that no part of this assessment or product has been submitted by me in another (previous or current) assessment, except where appropriately referenced, and with prior permission from the Lecturer for this unit.
- No part of the assessment/product has been written/produced for me by any other person except where collaboration has been authorised by the Lecturer concerned.
- **I am aware that this work will be reproduced and submitted to plagiarism detection software programs for the purpose of detecting possible plagiarism (which may retain a copy on its database for future plagiarism checking)**

Student's signature:	Yubaraj Thakulla	Nicholas Anthony De Leon	Sangay Yuden	Nayan Giri
Note: An examiner or lecturer / tutor has the right to not mark this assignment if the above declaration has not been signed.				

Table of contents

Ethical Security Considerations, User Interface Design.....	1
System Architecture.....	2
User Interface Design	5
Database Design.....	19
Entity Relationship Diagram.....	20
Key Entity relationships	21
Usability Testing and User Feedback	22
Designs before feedback and after feedback.....	24
Ethical Security Considerations	27
a. Ethical Issues	28
b. Security Threat Assessment.....	28
Conclusion	30

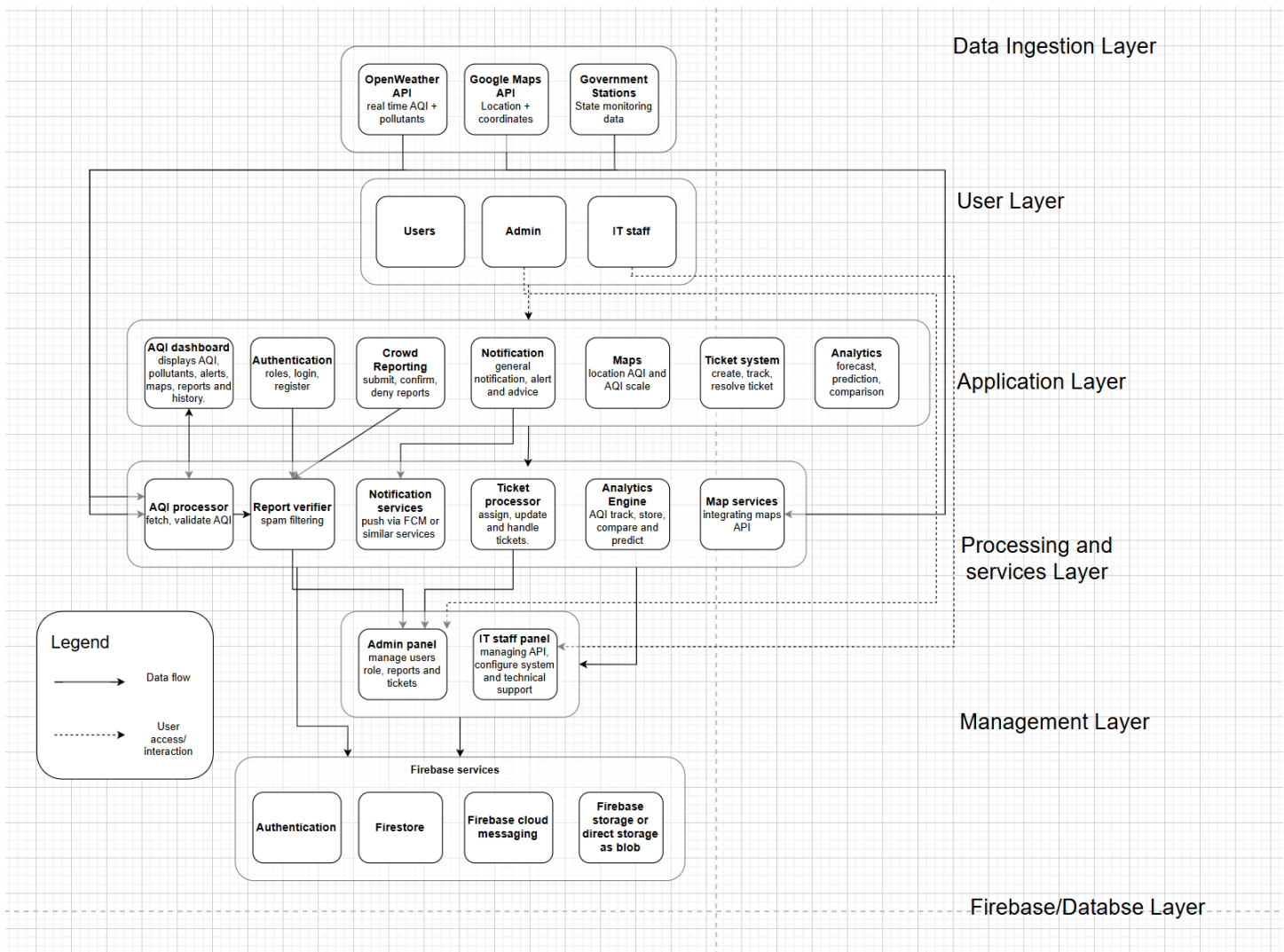
System Architecture

The air quality monitoring system is a proper project, so it needs a strong foundation so that the future steps can be proceeded with less conflict and smoothly. For that reason, a strong architecture design is needed. This need for architecture arises the architecture design for the air quality monitoring system. This system uses external APIs, needs a database, and has a lot of processes that have to be done in the backend. So, a multi-layered design is chosen to accommodate all the needs. The system is divided into data ingestion layer, user layer, application layer, processing and service layer, management layer, and firebase services layer which has database. Each layer has its own role: data ingestion, user access, application features, backend processing, and firebase database. This separation divides the system into smaller parts, making it organized and easier to build and maintain.

The system pulls the real-time data from external APIs, and those data will be fed to the AQI processor, which validates and collects data before displaying it to the users. All the stakeholders are shortened into three users, and they can access their own part of the system like users can check AQI, submit reports, and receive alerts, admins can manage users and reports, and IT staff can configure system and provide technical support. Frontend and backend are mainly divided into application and processing layers, where application layer contains everything that is visible to the users, and processing layer contains the logics to handle the system functions.

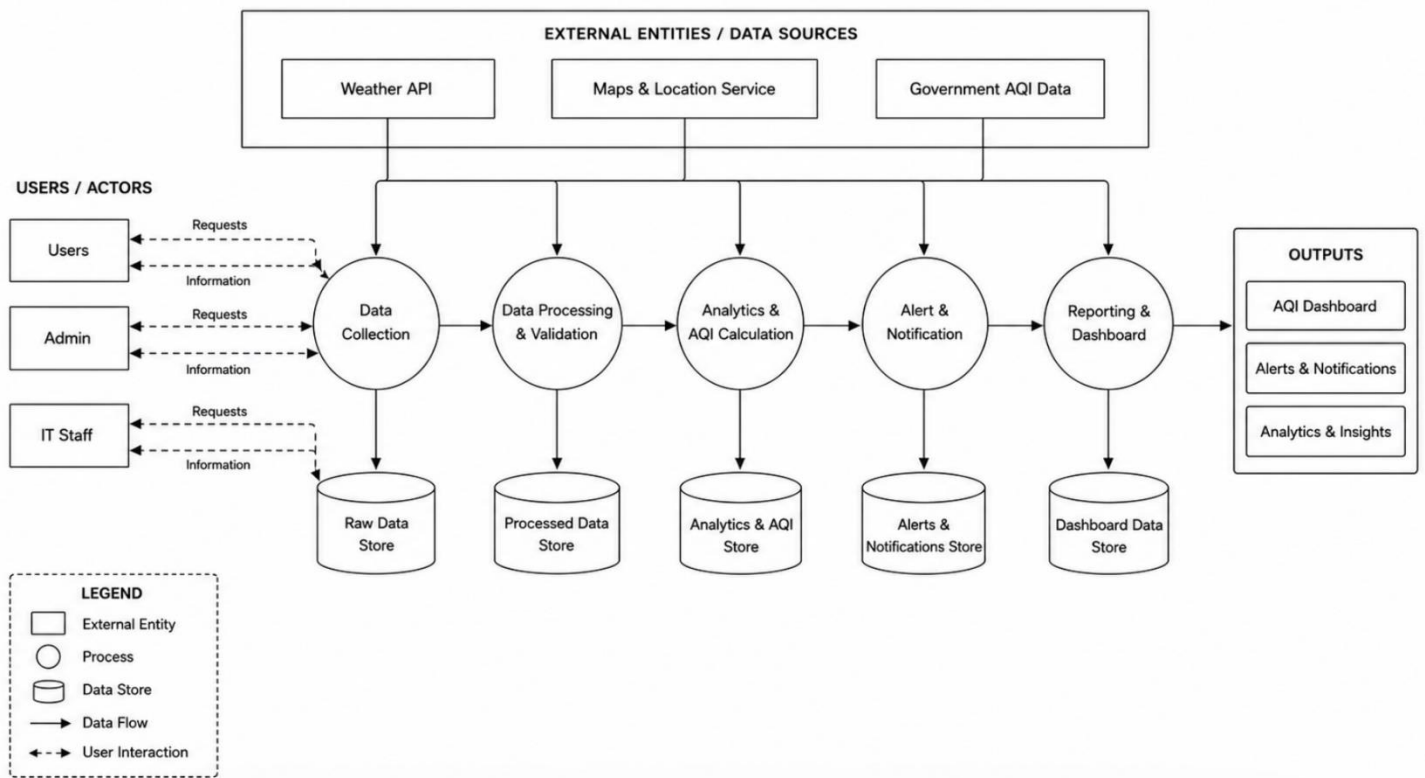
Previously it was mentioned that the system will use MySQL since that was the only database the team was familiar with for using with a system at that time. Since the system will be built in using flutter, and the team was taught firebase, firebase seems more reliable and has less things to worry about since it is cloud service. So, firebase will be used which has database, (Firebase cloud messaging) FCM for pushing the notifications, authentication for access management and storage, but the storage needs subscription, so all the data like images might be stored as a blob.

The flow is straight forward. The data from the external and internal sources will be stored in Firestore and compare them to safety thresholds. If a threshold is exceeded, an alert is created and distributed to users via FCM. When a user submits a crowd report, it is kept, compared against reports nearby, and shown to others so they can confirm or deny it. When it gets enough confirmations, it is marked as confirmed and users nearby are notified. Administrators and IT staff use the same Firestore database from their separate panels to manage data and system settings without requiring direct database access.



Architecture diagram

DATA FLOW DIAGRAM (DFD) – AIR QUALITY MONITORING SYSTEM



Data flow diagram

User Interface Design

The user interface of the application is kept simple and clean to make it easy to use by different users such as general users, vulnerable users, admin and IT staff. The application uses a bottom navigation bar with both icons and labels which is visible even while scrolling down or moving to different pages. The icons and labels help the user to easily differentiate between pages, and since it is always visible, it helps the user to move to different pages easily and quickly. The design uses soft, cool-toned colors to make it easy on the eyes. The background color is white to ensure that icons and other light colors stand out clearly. The text is kept large and readable. Overall, the user interface is designed to ensure ease of use by the users.

Screen	UI/UX Design Explanation
Sign Up and Login	The sign up and login screens are kept simple and clear, and users can enter their details, choose their role (general user/admin/IT), and access the system securely. The layout avoids unnecessary fields and uses clear buttons to reduce confusion.
User Dashboard	The dashboard gives users a quick overview of the current AQI, pollutant readings, alerts, and live reports. Large AQI numbers and color indicators help users to easily understand the air quality level quickly.
Map Page	The map page allows users to view AQI information by location. AQI bubbles and nearby area lists make it easier for the users to compare the air quality across different locations.
Reports Page	The reports page supports conversational reporting whereby users are allowed to submit and view live reports and confirm or deny them. This makes the reporting process simple.
Notification Page	The notification page displays AQI alerts, daily summaries, and health advice. This helps users respond quickly to unsafe air quality.
Admin Dashboard	The admin dashboard provides quick access to user and report management. It shows summary cards such as total users, active reports, and pending reviews so the admin can monitor the system efficiently.
IT Dashboard and System Configuration	The IT dashboard allows IT staff to monitor system status, open tickets, and configure settings such as API keys, AQI thresholds, and notification rules. This supports system maintainability and technical support.

1. Sign In and Login



Sign Up

Choose your role and create an account

Select Your Role

**User**
Standard user access to monitor air quality

**Admin**
Full administrative access and user management

**IT Support**
Technical access for system maintenance

Full Name

Email Address

Password

Confirm Password

Create Account

Already have an account? [Sign in](#)



Air Quality

Monitor air quality in real-time

Welcome back

Sign in to continue monitoring

Email Address

Password

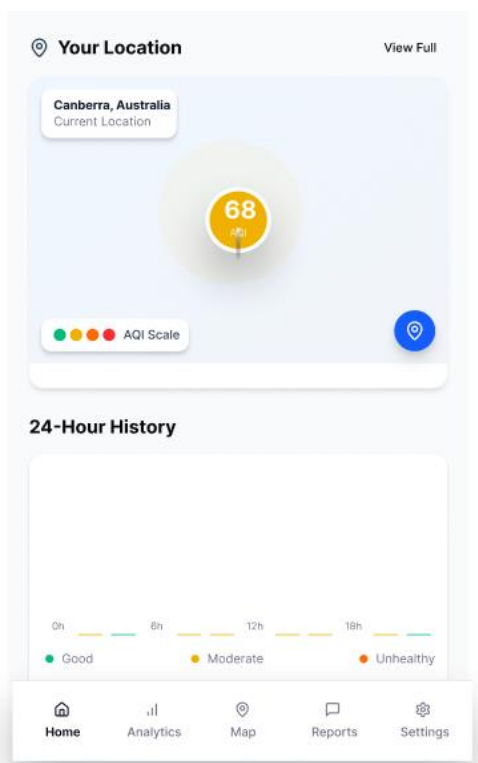
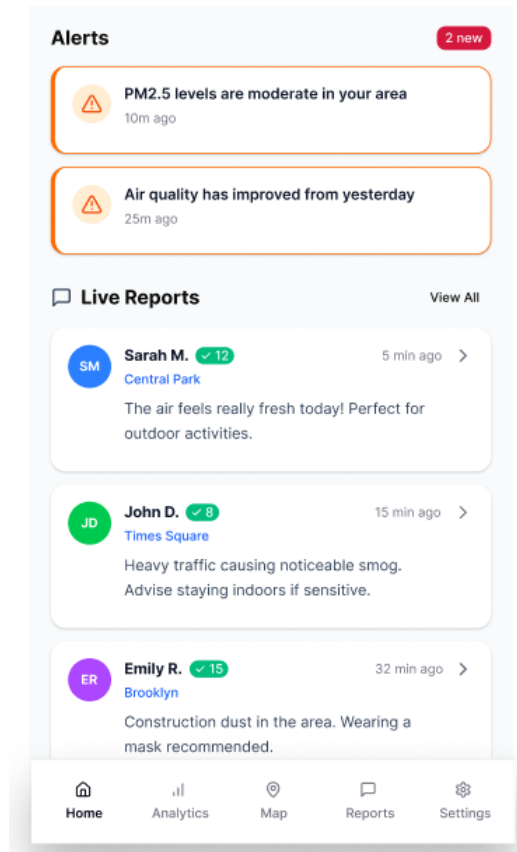
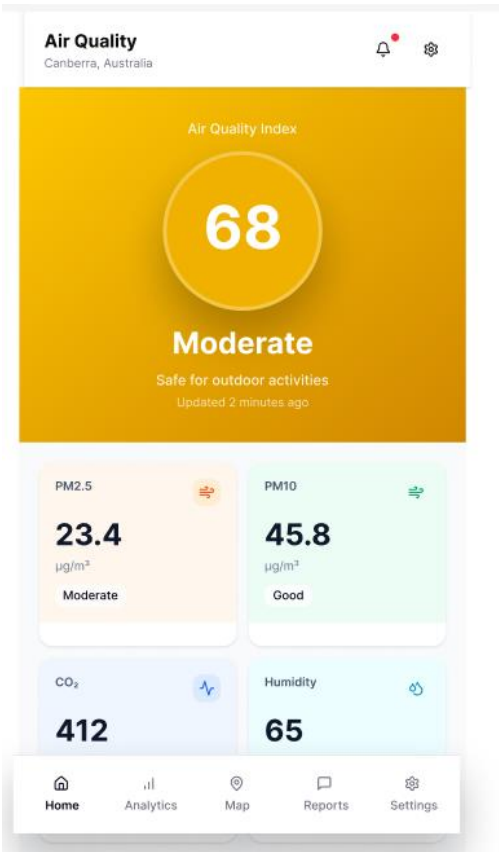
[Forgot?](#)

Sign in

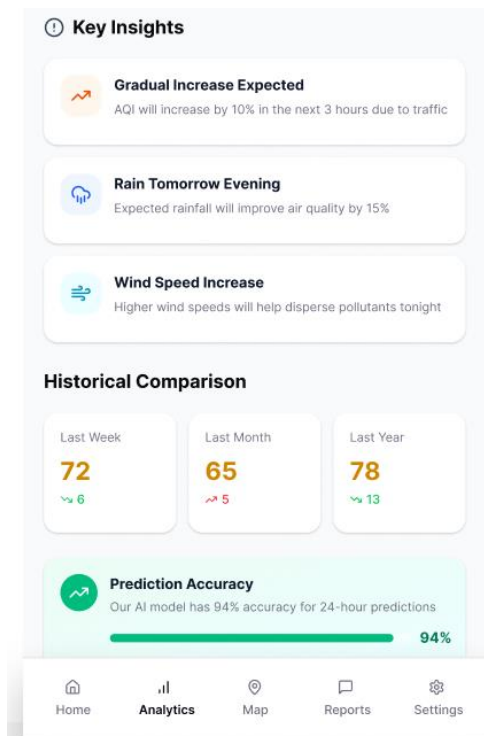
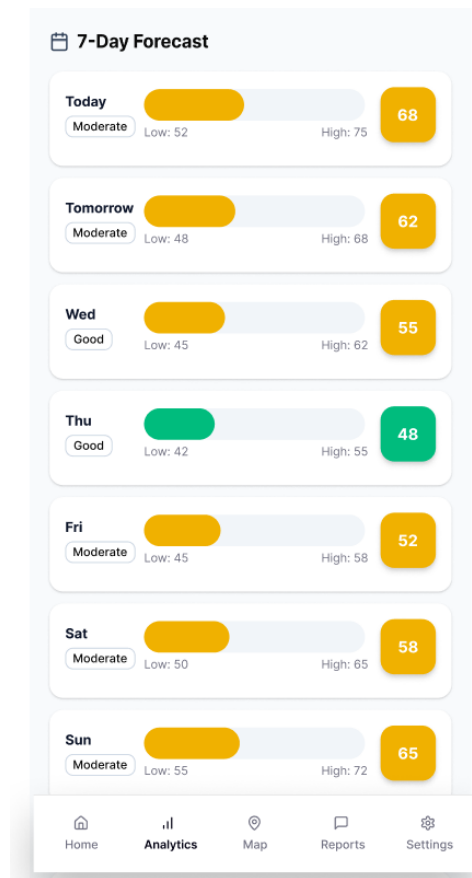
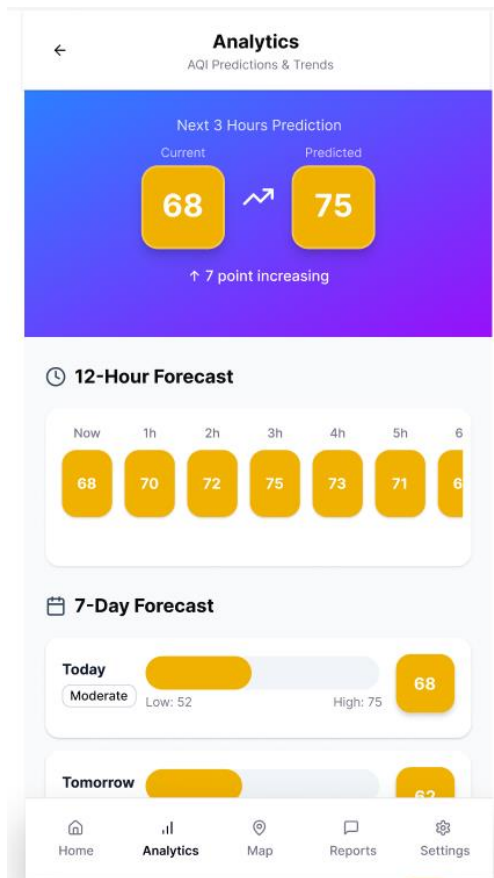
Don't have an account? [Sign up](#)

[Register with Role Selection →](#)

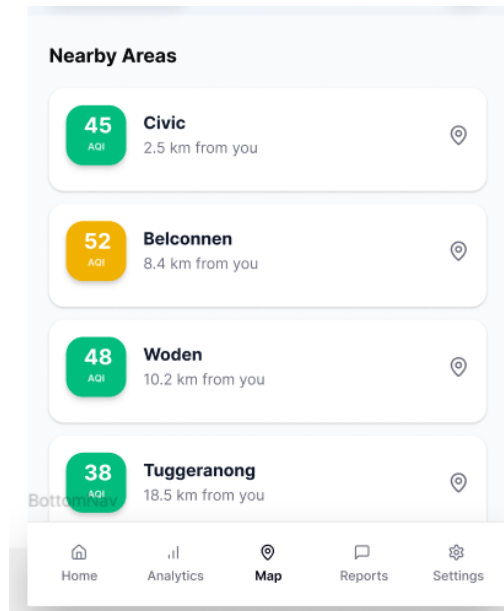
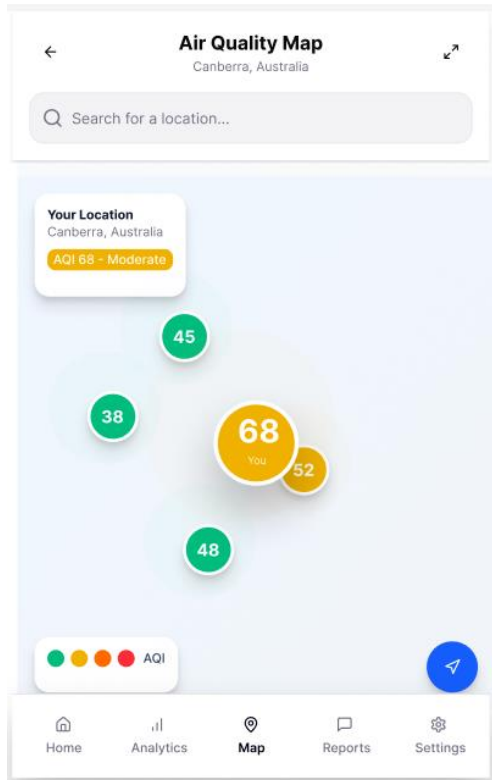
2. User Dashboard



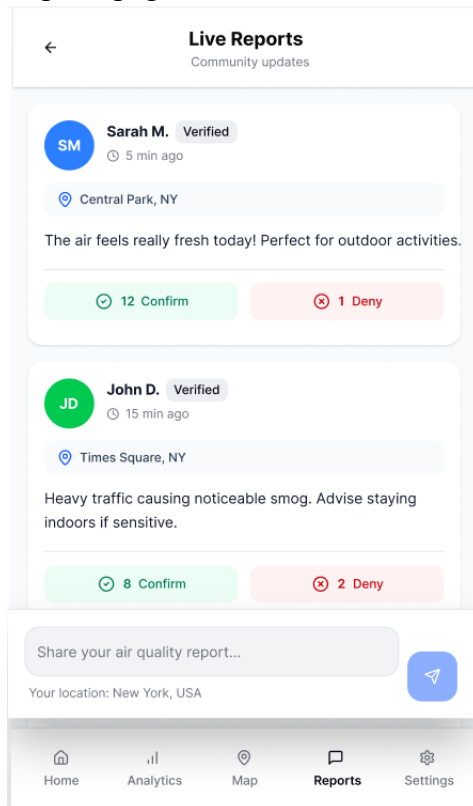
3. Analytics page



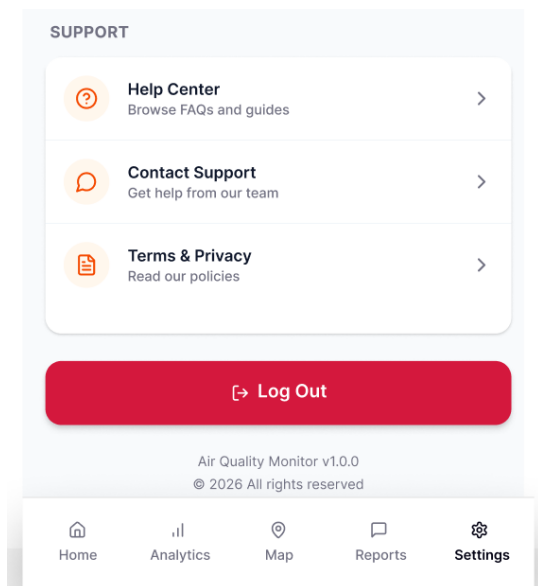
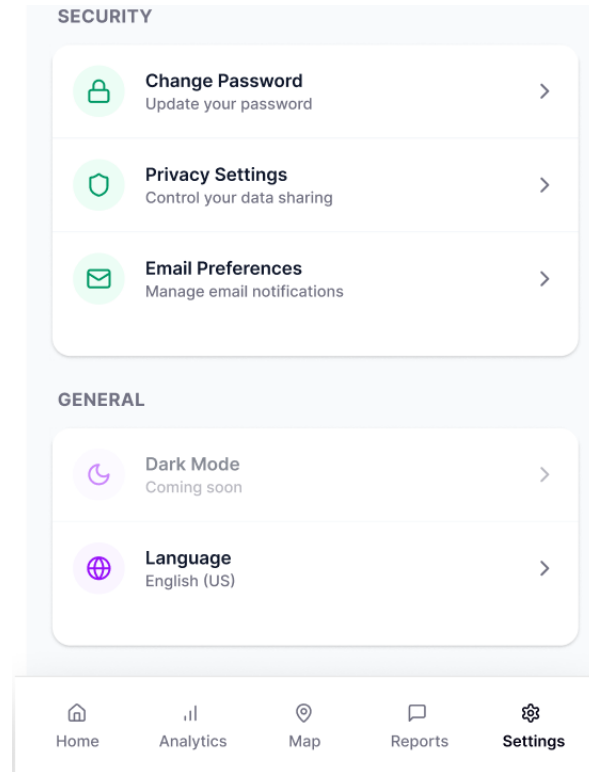
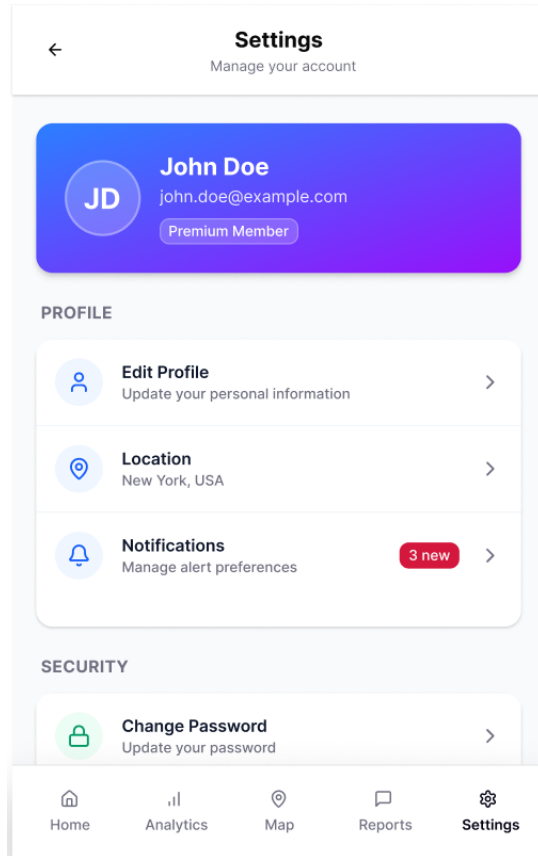
4. Air Quality Map page



5. Reports page



6. User Settings



7. Contact support page



Report a Problem

We're here to help



Describe your issue in detail and our support team will assist you as soon as possible.

Select Category *



Technical Issue



Data Accuracy



Feature Request



General Inquiry

Subject *

Brief description of the issue

Home

Analytics

Map

Reports

Settings

Description *

Please provide detailed information about your issue...

0/500 characters

Submit Ticket

By submitting, you agree to our support terms

Recent Tickets

TKT-482910

App crashes on startup

3 days ago

Resolved

TKT-471829

Incorrect AQI readings

1 week ago

In Progress

Home

Analytics

Map

Reports

Settings

8. Notification page



Notifications

2 unread notifications

All 2

Alerts

Health



Air Quality Alert

PM2.5 levels are moderate in your area

10 minutes ago



Daily Summary

Air quality has improved from yesterday

2 hours ago



Health Advice

Consider wearing a mask during outdoor activities

3 hours ago



High Pollution Warning

AQI levels expected to rise in the evening

Home

Analytics

Map

Reports

Settings

9. Admin Dashboard

Admin Access

Search users, reports, locations...

+12.5%

Total Users
12,458

+8.2%

Active Reports
342

-5.1%

Pending Reviews
23

Pending Reports
 View All

Sarah M. high pending

Bus Interchange, Civic
 Heavy smoke smell near the Civic bus interchange — came on suddenly
 5 min ago

James H. medium pending

Westfield Belconnen
 Construction dust causing discomfort around Belconnen shopping area
 12 min ago

Emma K. low approved

Woden Town Centre
 Air quality noticeably clean today — great for outdoor exercise
 23 min ago

Home

Users

Reports

Settings

Pending Reports
 View All

Sarah M. high pending

Bus Interchange, Civic
 Heavy smoke smell near the Civic bus interchange — came on suddenly
 5 min ago

James H. medium pending

Westfield Belconnen
 Construction dust causing discomfort around Belconnen shopping area
 12 min ago

Emma K. low approved

Woden Town Centre
 Air quality noticeably clean today — great for outdoor exercise
 23 min ago

Home

Users

Reports

Settings

Recent Users
 View All

Name	Email	Joined
Alice Johnson	alice.j@act.gov.au	2 hours ago
Bob Smith	bob.smith@canberrahealth.gov.au	5 hours ago
Carol White	carol.w@example.com.au	1 day ago
David Brown	david.b@example.com.au	2 days ago

Home

Users

Reports

Settings

10. Users page

Manage Users

6 users

Admin Access

Search users by name, email, or role...

6

Total Users

4

Active

2

Disabled

AJ

Alice Johnson

active

alice@example.com

user

Joined: 2 hours ago • Last active: 5 min ago

Edit Role

Disable

BS

Bob Smith

active

bob@example.com

admin

Joined: 5 hours ago • Last active: 1 hour ago

Edit Role

Disable

CW

Carol White

disabled

carol@example.com

user

Joined: 1 day ago • Last active: 2 days ago

Edit Role

Enable

DB

David Brown

active

david@example.com

it

Joined: 2 days ago • Last active: 10 min ago

Edit Role

Disable

ED

Emily Davis

active

emily@example.com

user

Joined: 3 days ago • Last active: 30 min ago

Edit Role

Disable

Home

Users

Reports

Settings

Home

Users

Reports

Settings

11. Reports page

Manage Reports

6 total reports

Admin Access

All (6)

Visible (5)

Hidden (1)

SM

Sarah M.

high

Visible

5 min ago

Bus Interchange, Civ

Smoke

Heavy smoke smell near the Civic bus interchange. Visibility reduced significantly this morning — came on suddenly around 8am.

12 confirm

1 deny

Community verified

Edit

Hide

JH

James H.

medium

Visible

15 min ago

Westfield Belconnen, Belconn

Dust

Construction dust has been really bad around the Belconnen shopping area today. Throat feels irritated and eyes are watering.

8 confirm

2 deny

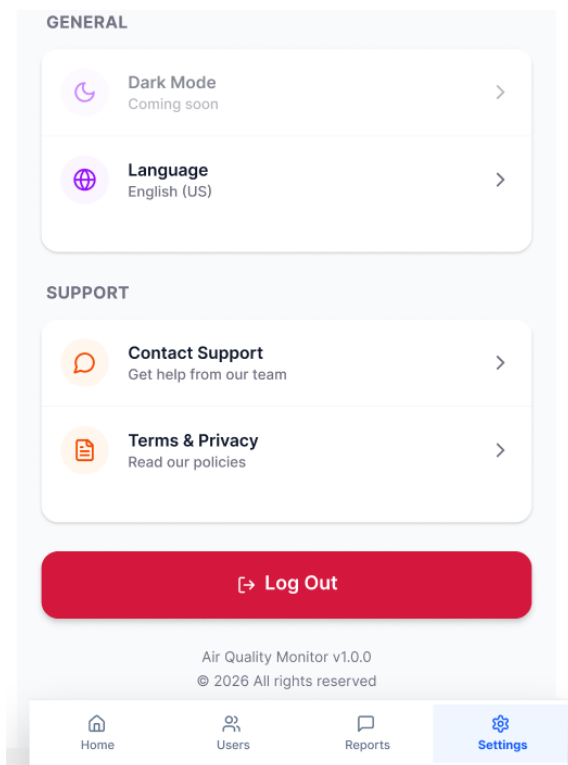
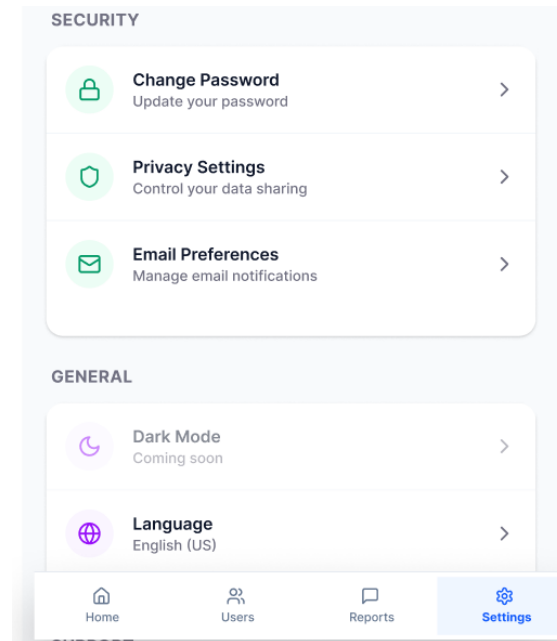
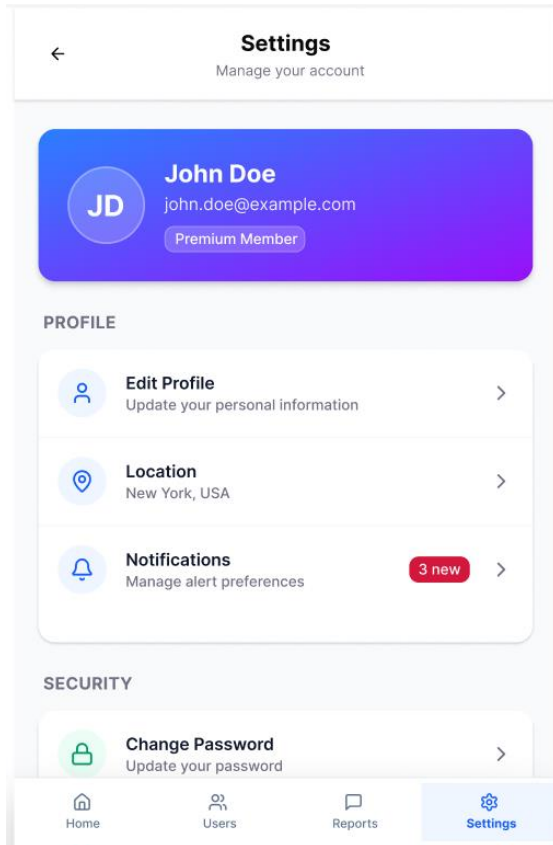
Home

Users

Reports

Settings

12. Admin Settings



13. IT Dashboard

IT Access

Search tickets, users, system logs...

+0.2%

System Uptime

99.8%

-3

Open Tickets

12

Stable

System Health

Good

Quick Actions

Home

Support

System

Settings

Quick Actions

System Config

Database

Security

Maintenance

Support Tickets

View All

Alice Johnson

high

open

Technical

Unable to access AQI data on mobile

15 min ago

Bob Smith

medium

in-progress

Account

Password reset request

1 hour ago

Home

Support

System

Settings

System Tasks

Database backup

Tonight 2:00 AM

scheduled

Security patch deployment

Tomorrow

pending

Server maintenance

Yesterday

completed

API rate limit review

This week

pending

System Alerts

API response time increased by 15%

30 min ago

Scheduled maintenance window in 6 hours

1 hour ago

Home

Support

System

Settings

Crown Institute of Higher Education

14. Technical support page

< ? Technical Support

Q Search tickets...

All

Open

In Progress

Resolved

#1 - Alice Johnson high

Unable to access AQI data on mobile

🕒 15 min ago open

#2 - Bob Smith medium

Password reset request

🕒 1 hour ago in-progress

#3 - Carol White high

Map not loading properly

🕒 2 hours ago open

#4 - David Brown low

Home

Support

System

Settings

15. System configuration page

< ☰ System Configuration

🔑 API Settings

API Key

sk_live_12345abcdef67890

API Endpoint URL

https://api.airquality.gov.au/v1

Timeout (seconds)

30

Max Retries

3

💾 Save API Settings

⚙️ AQI Threshold Settings

Good (0 - ?)

50

Moderate (50 - ?)

Home

Support

System

Settings

⚙️ AQI Threshold Settings

Good (0 - ?)

50

Moderate (50 - ?)

100

Unhealthy for Sensitive (100 - ?)

150

Unhealthy (150 - ?)

200

Very Unhealthy (200 - ?)

300

⚠️ AQI values above 300 are considered "Hazardous"

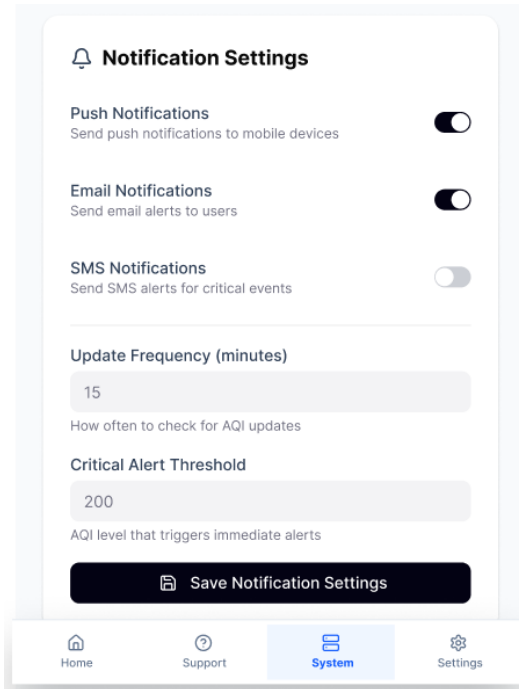
💾 Save AQI Thresholds

Home

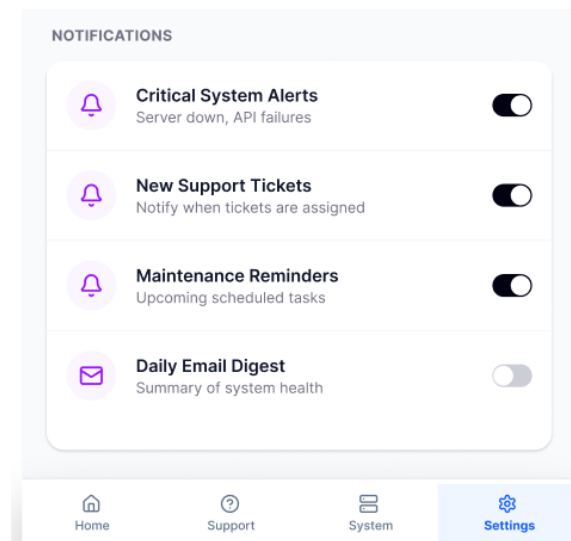
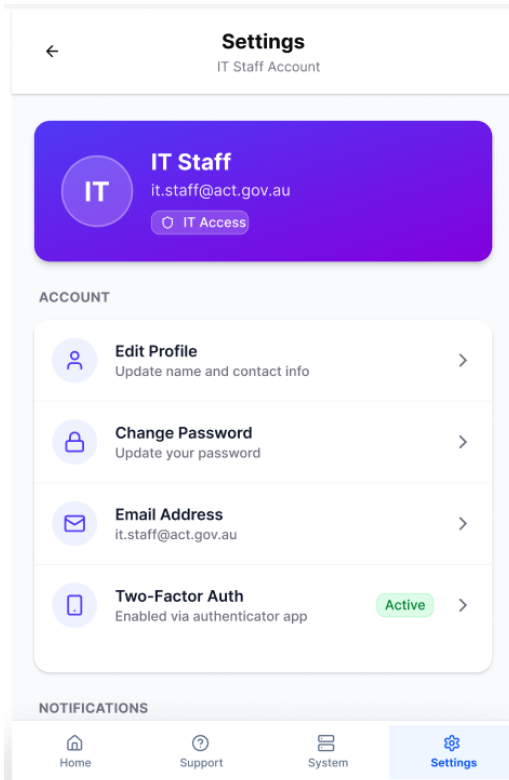
Support

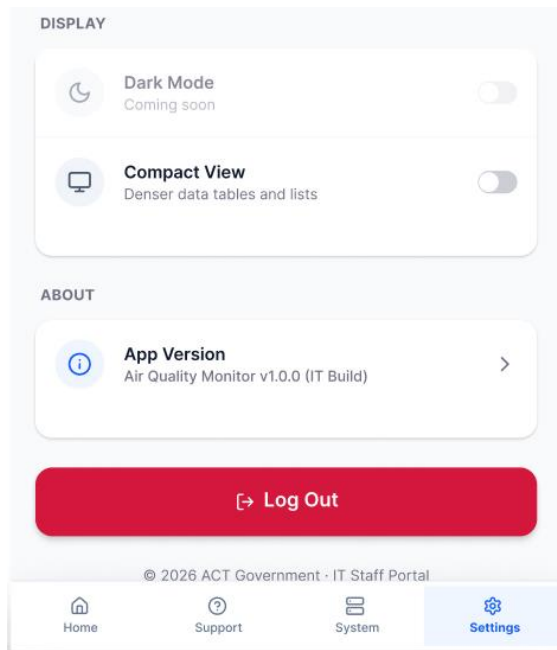
System

Settings



16. IT settings page





Database Design

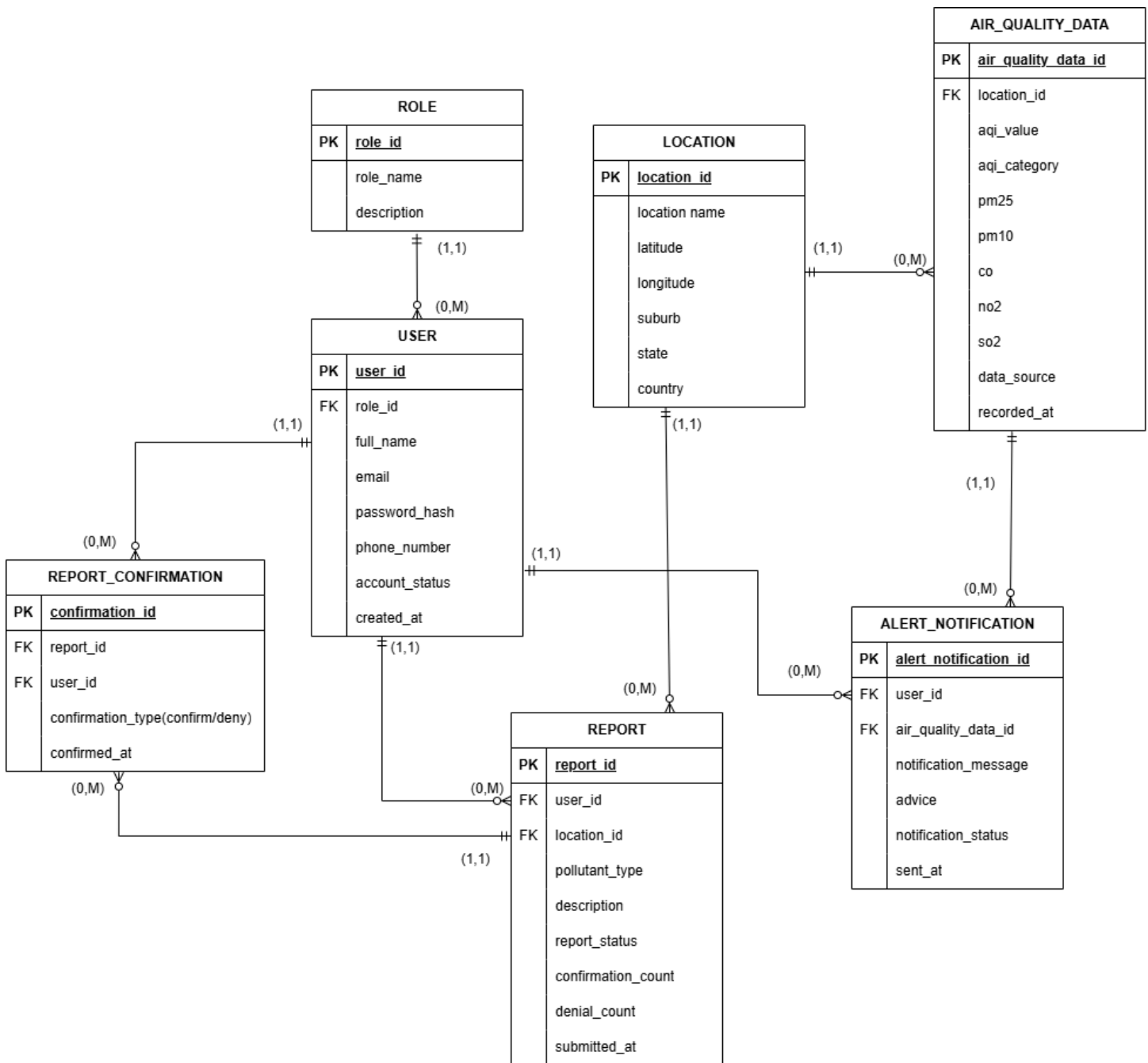
Entity Relationship Diagram

The ERD contains key entities that support the main functions of the system.

The key entities are :

1. Role
2. User
3. Locations
4. Air quality data
5. Reports
6. Report confirmations
7. Alert notification

The user entity stores user id and user details and inherit role id from role entity which helps to differentiate roles such as Admin, General user, and IT Staff. User entity also stores passwords in the form of hash to improve security. The location entity stores geological information and address. It is connected to both quality data and report entity because AQI and user reports need to have a specific location. The air quality data entity stores AQI information that is retrieved from external API. The report entity stores air quality reports submitted by the users whereas report confirmation entity stores user responses to reports such as confirm and deny. It is kept separate from the report entity because one report can receive many confirm or deny from many different users. This avoids repeated data. The alert notification entity stores alerts, notifications, and advice messages that are sent to the users. It is connected to the air quality data entity because notifications are triggered by the AQI records when the air quality level is unsafe. Overall, the ERD shows how the main entities are stored and connected to each other.



This is an Entity Relationship Diagram

Key Entity relationships

No	Relationship	Relationship Type	Explanation
1	Role - User	1,1 to 0, M	Each user must have exactly one role, such as General User, Admin, or IT Staff. One role can be assigned to zero or many users.
2	User - Report	1,1 to 0, M	One user can submit zero or many reports. Each report belongs to one user.
3	Location - Report	1,1 to 0, M	One location can have zero or many user reports. Each report must be linked to one specific location.
4	Location - AirQualityData	1,1 to 0, M	One location can have many air quality records over time. Each air quality data record belongs to exactly one location.
5	Report – Report_Confirmation	1,1 to 0, M	One report can receive zero or many confirmations or denials. Each confirmation belongs to exactly one report.
6	User - Report_Confirmation	1,1 to 0, M	Many users can confirm or deny many reports. This many-to-many relationship is resolved using the Report_Confirmation entity. Where one user can confirm or deny zero or many reports. Each confirmation is made by exactly one user.
7	User - Alert_Notification	1,1 to 0, M	One user can receive zero or many notifications.
8	Air_Quality_Data – Alert_Notification	1,1 to 0, M	One AQI record can trigger many alert notifications.

Usability Testing and User Feedback

Usability testing was conducted in order to collect feedback from the users about the system. The main purpose of the test was to check if the system was easy to use, or complex, and whether the user was willing to accept the application.

The test was conducted both in person and remotely using Microsoft forms. The form contained SUS questionnaire. It included 5 participants. Some participants were asked to sign up and login as general users, some as admin, and some as IT Staff. The participants were required to view the dashboard and navigate through the bottom navigation bar.

General users were required to check the different features such as report submission, confirm or deny, and ticket submissions. They were also asked to view notifications and check if they can navigate to respective pages by clicking. They were also required to check if the map was working and whether they could check location specific data.

The admin users were required to check if they could easily manage user roles and live reports. And IT staff were required to check if they could check the submitted reports and provide support. They were also required to check if they could easily check the system configuration and save them.

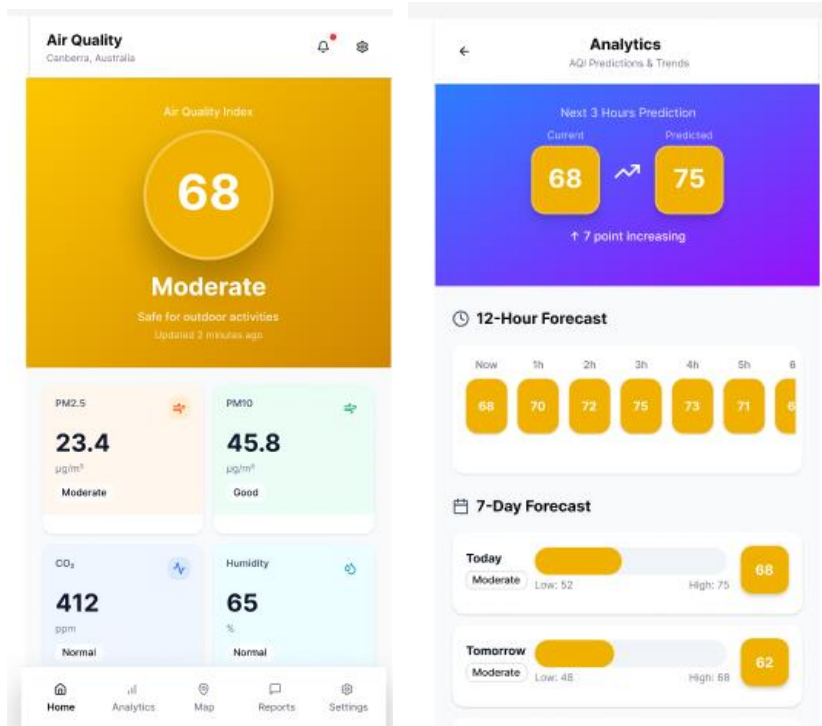
Feedback to Action Table

User Feedback	Issue Type	Severity	Design Change Made	Result/ Validation
“The navigation bar was not visible in all the pages”	Navigation	Critical	Made sure the navigation bar was visible at the bottom in all the pages.	Navigation bar was made consistent throughout all the pages.
“The ticket submission is accessible from the user dashboard which makes the dashboard cluttered and confusing.”	Design error	Major	The ticket submission was moved to settings/support.	The dashboard focused only on AQI information.
“The dashboard screen for Admin and IT were in computer screen format”	Design error	Major	Made the layout was made consistent for all the users.	The layout for all the users was made within the scope i.e followed mobile layout

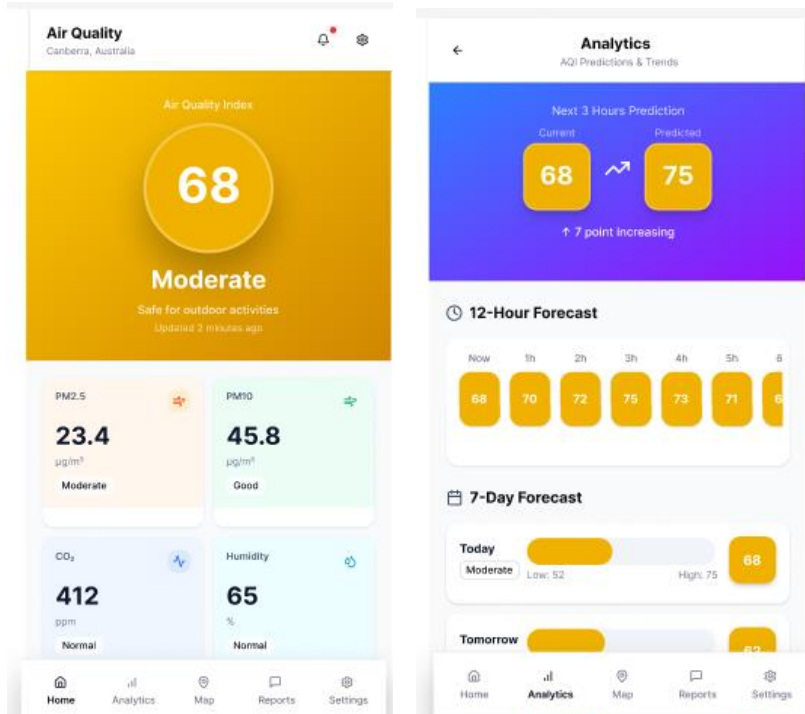
Multiple users gave the same feedback but in their own way, so even though there were more users, we got only three feedback.

Designs before feedback and after feedback

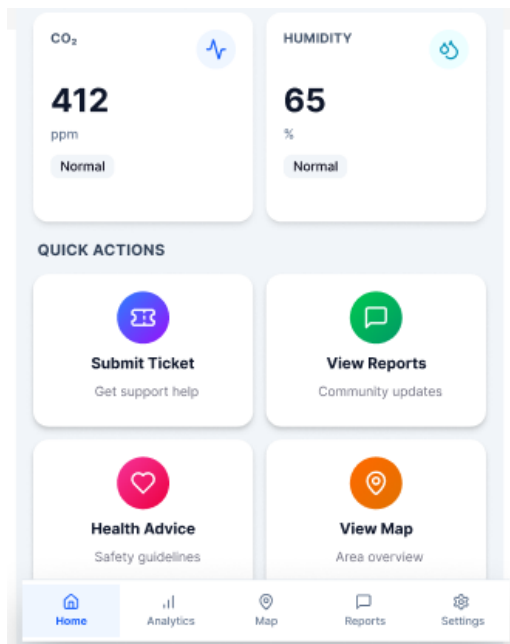
1. Inconsistent navigation



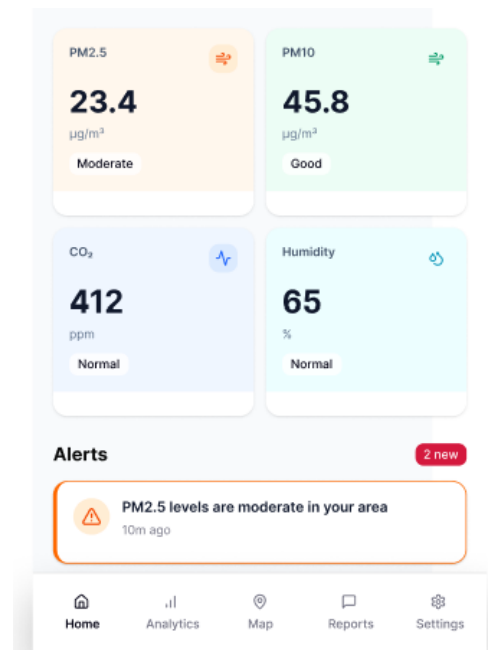
Consistent navigation After feedback



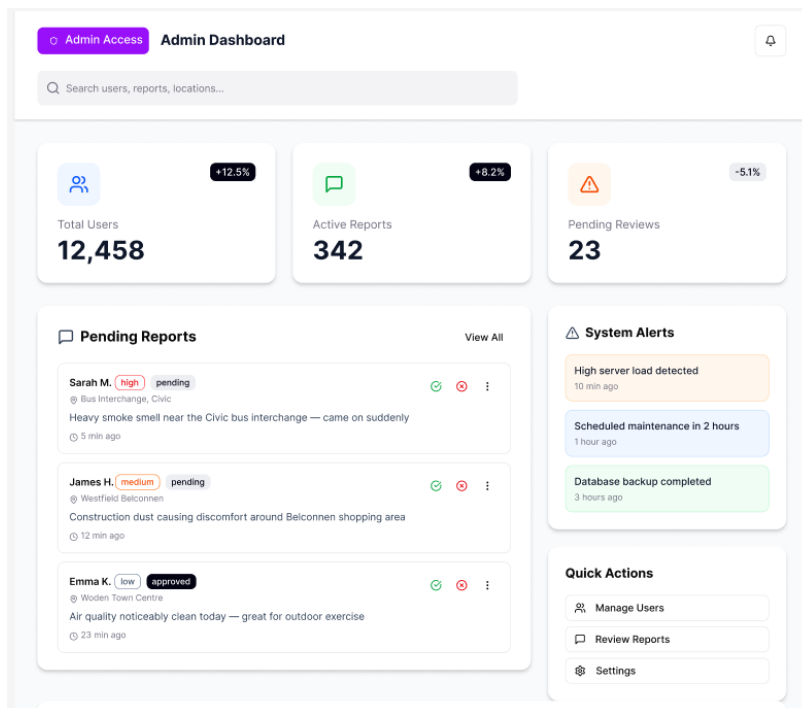
2. Ticket submission in dashboard



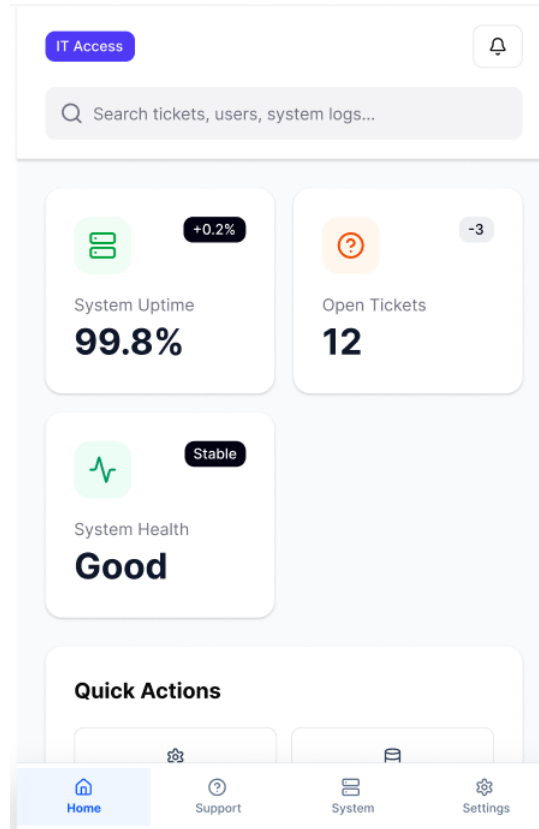
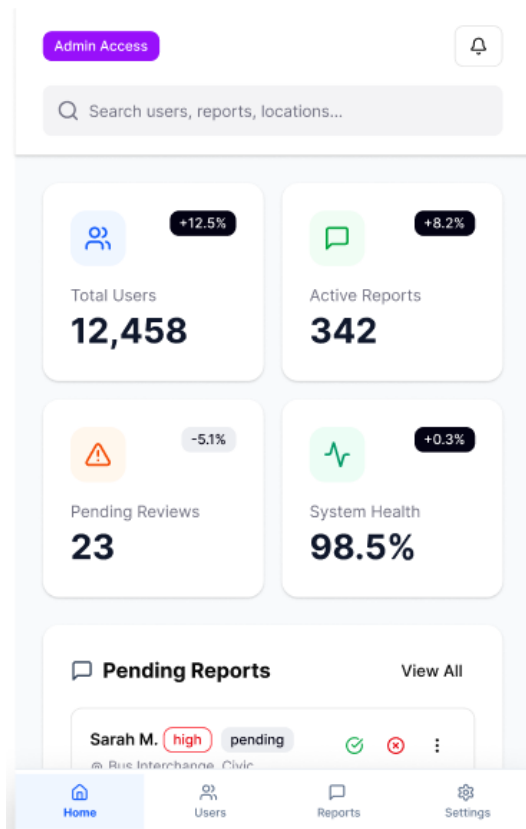
Dashboard after removing ticket submission



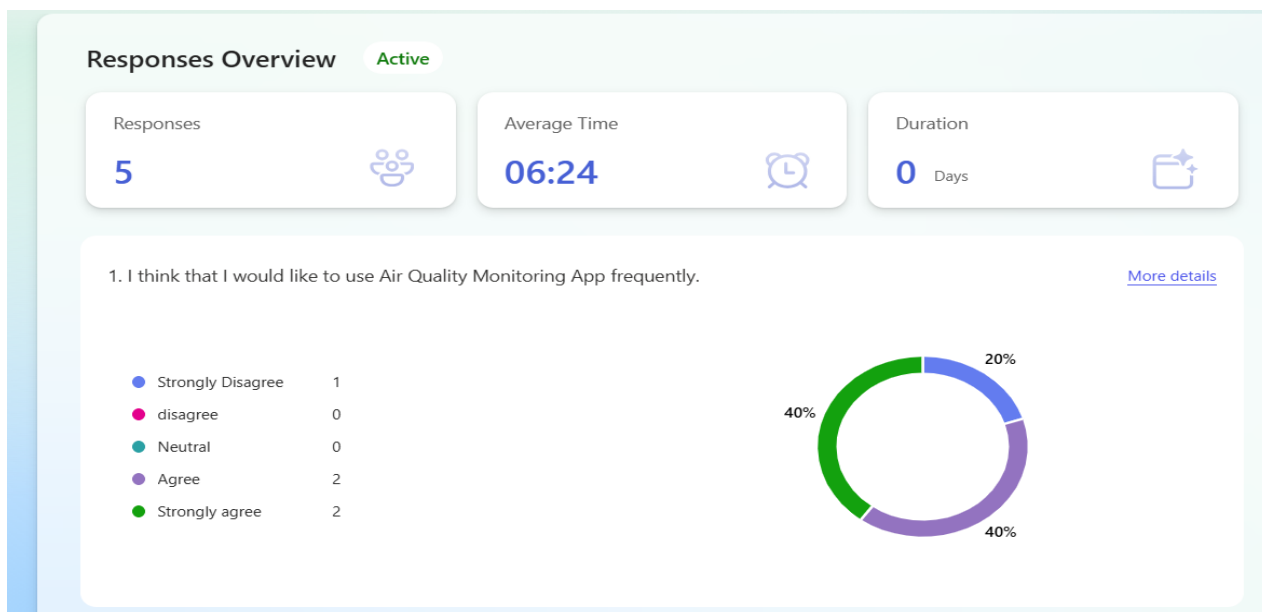
3. Inconsistent Admin and IT dashboard with user dashboard



After making it consistent with user dashboard



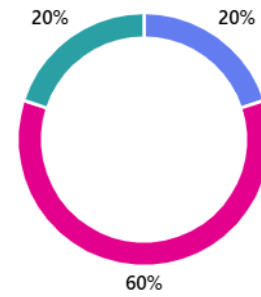
Microsoft SUS Survey Form



2. I found the app unnecessarily complex.

[More details](#)

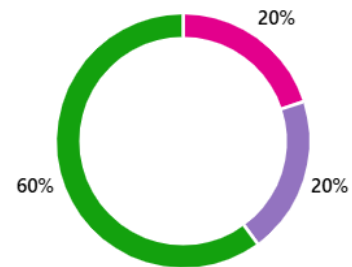
Strongly disagree	1
Disagree	3
Neutral	1
Agree	0
Strongly agree	0



3. I thought the app was easy to use.

[More details](#)

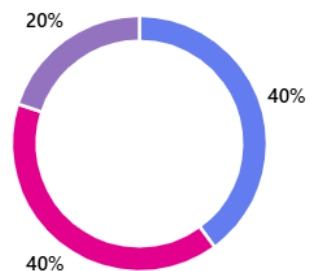
Strongly disagree	0
Disagree	1
Neutral	0
Agree	1
Strongly agree	3



4. I think that I would need the support of a technical person to be able to use the app.

[More details](#)

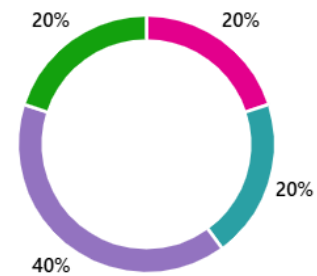
Strongly Disagree	2
Disagree	2
Neutral	0
Agree	1
Strongly agree	0



5. I found the various functions in app were well integrated.

[More details](#)

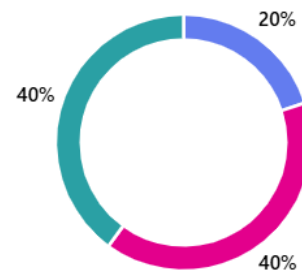
Strongly disagree	0
Disagree	1
Neutral	1
Agree	2
Strongly agree	1



6. I thought there was too much inconsistency in your app.

[More details](#)

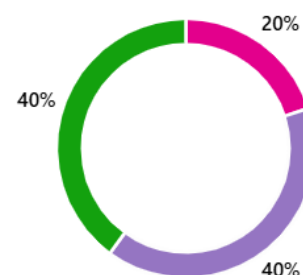
Strongly Disagree	1
Disagree	2
Neutral	2
Agree	0
Strongly agree	0



7. I would imagine that most people would learn to use the app very quickly.

[More details](#)

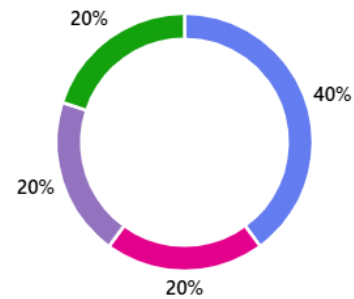
Strongly disagree	0
Disagree	1
Neutral	0
Agree	2
Strongly agree	2



8. I found the app very cumbersome to use.

[More details](#)

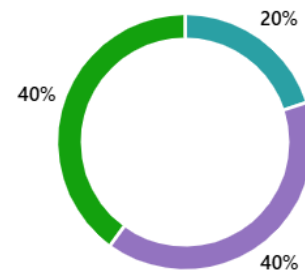
Strongly disagree	2
Disagree	1
Neutral	0
Agree	1
Strongly agree	1



9. I felt very confident using the app

[More details](#)

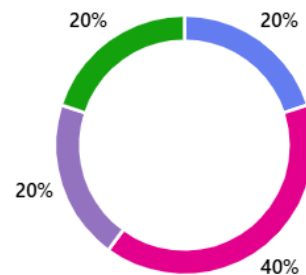
Strongly Disagree	0
Disagree	0
Neutral	1
Agree	2
Strongly Agree	2



10. I needed to learn a lot of things before I could get going with the app.

[More details](#)

Strongly disagree	1
Disagree	2
Neutral	0
Agree	1
Strongly agree	1



Usability Testing – SUS Score Results

Participant	Odd Total	Even Total	$X = \text{Odd} - 5$	$Y = 25 - \text{Even}$	SUS Score
Participant 1	18	12	13	13	65.0
Participant 2	23	16	18	9	67.5
Participant 3	22	8	17	17	85.0
Participant 4	11	17	6	8	35.0
Participant 5	25	5	20	20	100.0

Average SUS Score: 70.5

Ethical Security Considerations

a. Ethical Issues

- **Opaque mixing of API and user reports (Low)**
 - o Users may misinterpret the combined report of API and user reports.
- **Unclear confidence and validation of hazards (Medium)**
 - o Waze-style confirmations can be mistaken for “truth”, and people with health vulnerabilities might rely on uncertain reports as oversimplified indicators.
- **Ambiguous status of health guidance (High)**
 - o If the app shows health tips (i.e. “Limit outdoor activity”) without clarifying whether they are general guidance or specific advice for select users, people may assume a level of medical authority that is not actually present.
- **Location tracking (High)**
 - o Since hazard reports are inherently location-based and time-stamped, repeated use can reveal sensitive patterns like home, work, or school, even if names are hidden.
- **Inadequate consent and control (Medium)**
 - o Users may consent once when the app is installed but are not reminded that their location collection may happen in the background.

- **Location and identity exposure through data breaches (High)**
 - o Data breaches could reveal logs of where and when users reported hazards and expose sensitive data.
- **Unequal coverage (Low)**
 - o Crowdsourced reporting tends to overrepresent urban areas instead of rural areas where smartphone use and digital literacy is higher.

b. Security Threat Assessment

Threat	Risk Level	Why It's Relevant	Mitigation Strategies	Affected Components
Flawed Authentication	Critical	Compromised accounts can submit false hazard/AQI reports, misleading health-vulnerable users into dangerous decisions.	Enforce MFA, use short-lived JWT with refresh rotation, crypt/Argon2 password hashing, and lock accounts after failed logins.	User auth service, hazard submission API, mobile login flow
Flawed Access Control	High	Without ownership checks, users could alter or delete other users' reports or access aggregated location data.	Server-side ownership checks on every endpoint, UUIDs to prevent ID enumeration, log all access control failures.	Report management API, database row-level permissions, admin dashboard
SQL / NoSQL Injection	High	Hazard description fields, location queries and filters are user-controlled; unsensitized inputs could expose or corrupt the entire database.	Use parameterized queries exclusively; apply an ORM with strict type validation; whitelist and sanitizes all input server-side.	Hazard submission form, location search, AQI filter queries, database layer
Sensitive Data Exposure	Critical	Time-stamped location reports reveal home/work patterns. Combined with health activity, this is among the most sensitive personal data categories.	Encrypt location data at rest (AES-256) and in transit (TLS 1.3+); apply data minimization; offer location fuzzing as a user option.	Location database, user profile store, hazard report logs, API responses
Cross-Site Scripting (XSS)	Medium	Hazard descriptions rendered in a web dashboard without escaping allow	Apply output encoding on all user-generated content; enforce a strict	Web dashboard, hazard report

		attackers to embed scripts that steal session tokens or redirect admins.	Content Security Policy; sanitizes rich text with an allowlist library (e.g. DOM Purify).	display, admin review interface
DDoS Attacks	High	AQI retrieval and hazard submission are publicly facing. An attack during a real emergency (e.g. bushfire smoke) denies access when the app matters most.	API gateway rate limiting and per-IP throttling; CDN/WAF for traffic scrubbing; auto-scaling on cloud infrastructure; maintain a status page.	AQI retrieval API, hazard submission endpoint, cloud hosting layer

Conclusion

The overall system structure was explained while also showing how system components interact with each other. Also included suitable diagrams. User interface design was briefly described with a table and all required screenshots. The Figma was used to create the UI/UX design. Database design was created with an ERD to show the database visually while also explaining the key entities and relationships in a table. Got feedback through the usability testing and changed the system accordingly. Ethical and privacy issues were identified, and mitigation strategies were mentioned accordingly.